Published in Geek Culture

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A Simple Guide to The (New) ChatGPT API with Python

ChatGPT API was just released. Explore everything you can do with the ChatGPT API in Python.

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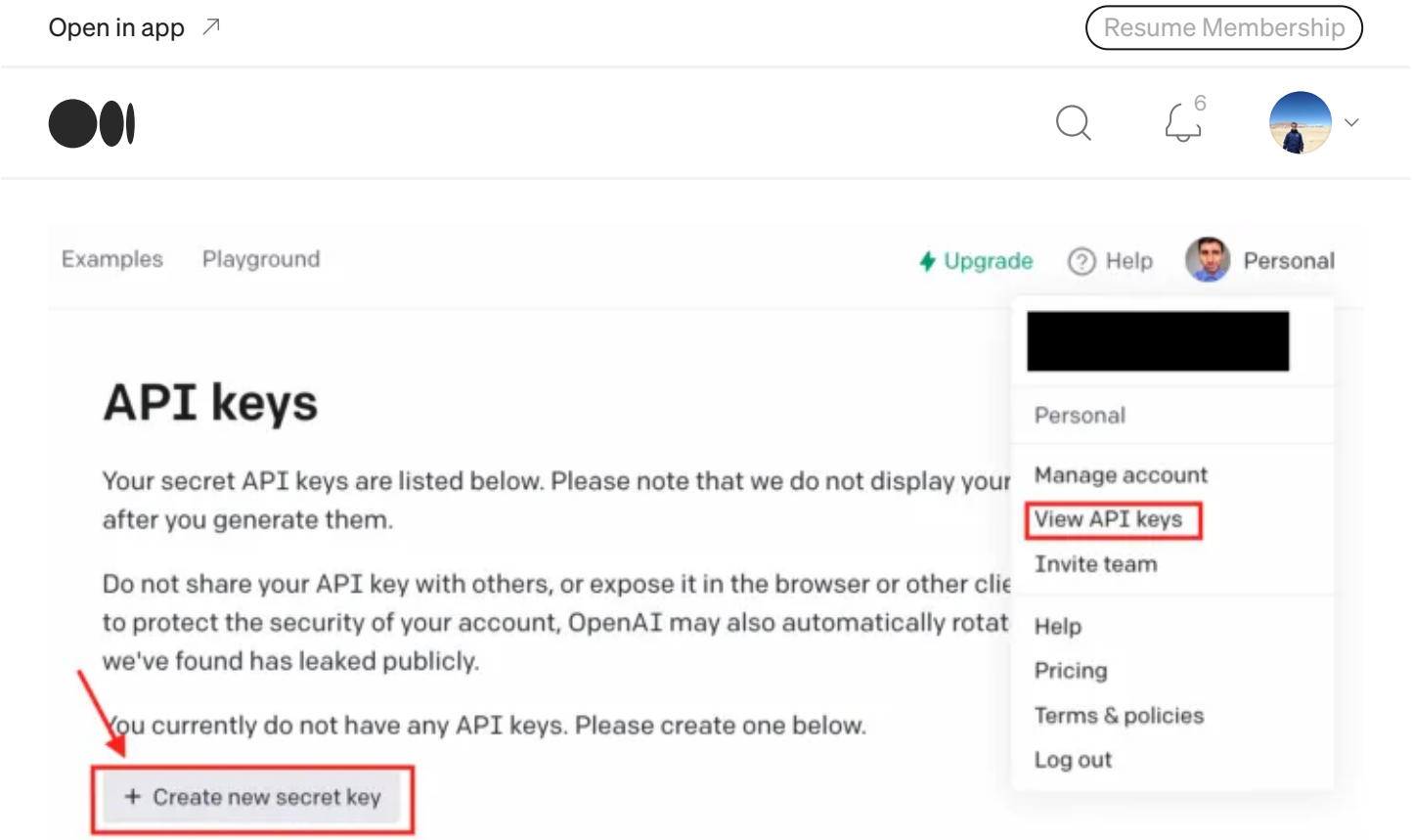
OpenAI just released ChatGPT API. This is an API that calls gpt-3.5-turbo, which is the same model used in the ChatGPT product.

For those already familiar with the OpenAI API in Python, learning how to use the ChatGPT API should be simple, but there are still some concepts that are exclusive to this API that we'll learn in this guide.

Let's explore the ChatGPT API in Python.

Note: The API is priced at $0.002 per 1K tokens. You have free credit to use, though. It seems now I have another reason to cancel my ChatGPT Plus subscription.



Remember that OpenAI won't display your secret API key again after you generate it, so copy your API key and save it. I'll create an environment variable named `OPENAI_API_KEY` that will contain my API key for this tutorial.

Installing the library

To work with the ChatGPT API, first, we have to install the openai library by running the following command.

```
pip install openai
```

There are a lot of things you can do with the OpenAI library, but today we're going to focus on the Completion endpoint. To be more precise, we'll use "ChatCompletion" `gpt-3.5-turbo`, which is the same model used by ChatGPT.

Note: This guide is also available in video format on my YouTube channel.

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Getting started with the ChatGPT

To get started, we're going to use the code snippet below that I got from the official documentation.

```
import os
import openai
openai.api_key = os.getenv("OPENAI_API_KEY")

completion = openai.ChatCompletion.create(
  model="gpt-3.5-turbo",
  messages=[
    {"role": "user", "content": "Tell the world about the ChatGPT API in the st
  ]
)

print(completion.choices[0].message.content)
```

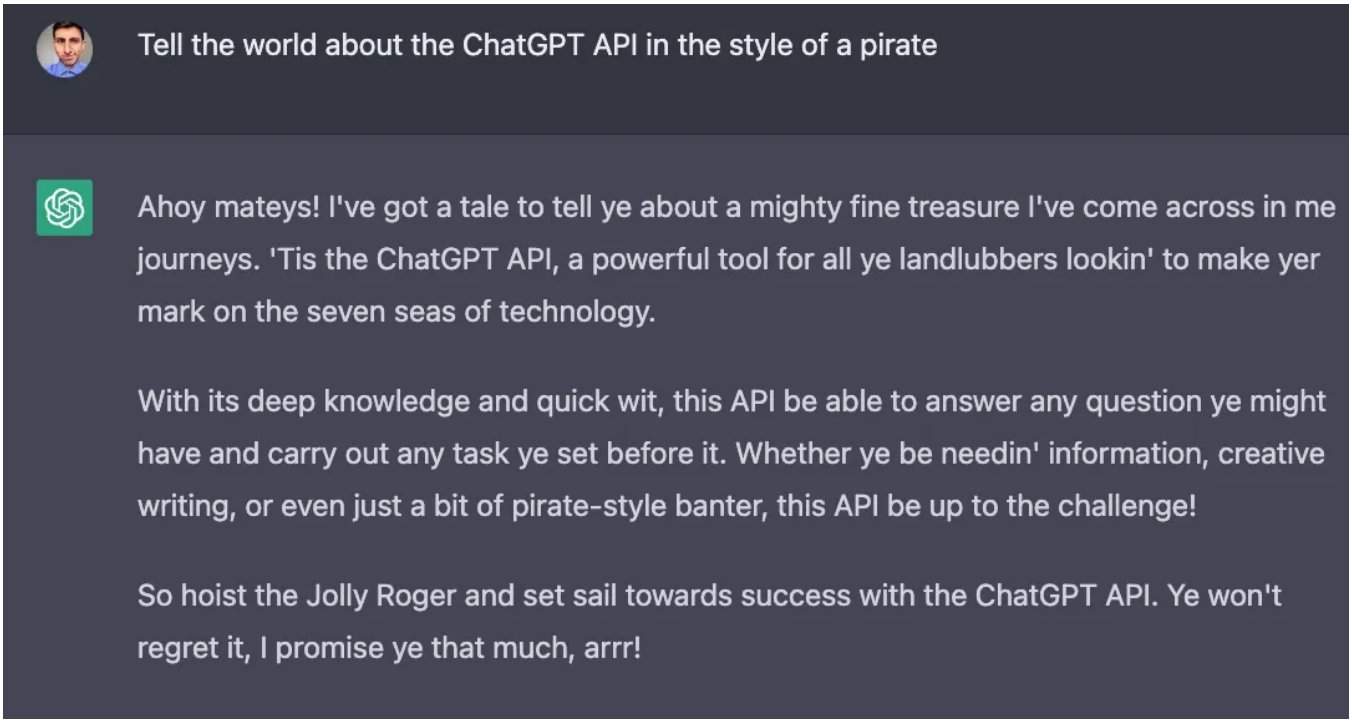
This is a very simple way to interact with the API. The main thing to know is that the `messages` list has a dictionary with 2 keys: `roles` and `content`.

The `content` is simply the content of the message, while there are three main `roles`: "system", "user", or "assistant". The "user" is the one who gives the instructions and is being used in the code above.

If we run the code above, we'll get the following.

> *Ahoy matey, ye be lookin' for a chat API that'll blow ye away? Look no further than ChatGPT API ye ol' sea dog!*
>
> *With ChatGPT API, ...*

This is the same as asking ChatGPT “Tell the world about the ChatGPT API in the style of a pirate.”



Now, the code snippet we took from the docs is very basic, we can add more code to interact with the API as if we were chatting with ChatGPT. Also, we should add the `system` role to set the behavior of the assistant and the `assistant` role to store prior responses.

System Role

Let’s add a system role to set the behavior of the assistant. You can tell the assistant to be a “helpful assistant that translates English to French” or anything you want.

To make it simple, I’ll set the system role to be a “kind helpful assistant” through the messages list (on my YouTube video, I made ChatGPT behave as a recruiter in a job interview)

```
messages = [
 {"role": "system", "content" : "You’re a kind helpful assistant"}
]
```

Now I’m going to append the `users` role to the previous list and add the input function in order to interact with the API as if we’re working with ChatGPT.

```python
import openai

content = input("User: ")
messages.append({"role": "user", "content": content})

completion = openai.ChatCompletion.create(
  model="gpt-3.5-turbo",
  messages=messages
)

chat_response = completion.choices[0].message.content
print(f'ChatGPT: {chat_response}')
```

Now if we ask “who was the first man on the moon?” the system will behave as a “helpful assistant and tells us the answer”

```
User: who was the first man on the moon?
ChatGPT: The first man on the moon was Neil Armstrong on July 20, 1969.
```

There’s only one little detail, though. The `assistant` isn’t storing the previous responses, so the system might not remember our previous responses and give us a proper response.

Let’s now ask “where is he from?” and then “how tall is he?” I will now add a while loop to ask multiple questions.

```
In [*]: import openai

        while True:
            content = input("User: ")
            messages.append({"role": "user", "content": content})

            completion = openai.ChatCompletion.create(
              model="gpt-3.5-turbo",
              messages=messages
            )

            chat_response = completion.choices[0].message.content
            print(f'ChatGPT: {chat_response}')

        User: who was the first man on the moon?
        ChatGPT: The first man on the moon was Neil Armstrong on July 20, 1969.
        User: where is he from?
        ChatGPT: The first man on the moon was Neil Armstrong. He was from Wapakoneta, Ohio in the United States.
        User: how tall is he?
        ChatGPT: The first man on the moon was Neil Armstrong. He was an American astronaut born in Wapakoneta, Ohio. He pa
        ssed away on August 25, 2012. Neil Armstrong was 5 feet 11 inches (1.80 meters) tall.

        User:
```

As you can see, now the system is answering each new question, but it’s accumulating the information from the previous replies.

We can solve this issue with the `assistant` role.

Assistant Role

We use the assistant role to store prior responses. By storing previous responses, we can build a conversation history that will come in handy when user instructions refer to prior messages.

With the code below, we add the assistant role to our messages list.

```
messages.append({"role": "assistant", "content": chat_response})
```

Note that in the content we have to add the chat_response in order to store the responses.

Now we add the previous line of code to our while loop.

```
import openai

while True:
    content = input("User: ")
    messages.append({"role": "user", "content": content})

    completion = openai.ChatCompletion.create(
      model="gpt-3.5-turbo",
      messages=messages
    )

    chat_response = completion.choices[0].message.content
    print(f'ChatGPT: {chat_response}')
    messages.append({"role": "assistant", "content": chat_response})
```

Now if we ask the same question we asked before, we'll get better responses.

```
User: who was the first man on the moon?
ChatGPT: The first man on the moon was Neil Armstrong, an American astronaut who made his
tory on July 20, 1969 when he became the first person to step onto the lunar surface as p
art of the Apollo 11 mission.
User: where is he from?
ChatGPT: Neil Armstrong was born on August 5, 1930, in the city of Wapakoneta, Ohio, Unit
ed States. So, he was an American astronaut.
User: how tall is he?
ChatGPT: Neil Armstrong was approximately 5 feet 11 inches (180 cm) tall.

User:
```

And that's it! Now you know how to work with the 3 roles, you can use the API as you want. Besides the model and messages in the request body, you can also add other parameters to customize your request.

Here are some extra parameters:

- `max_token` : The maximum number of tokens to generate in the completion (here you can see the tokenizer that OpenAI uses)
- `temperature` : The sampling temperature to use. Values close to 1 will give the model more risk/creativity, while values close to 0 will generate well-defined answers.
- `n` : The number of chat completion choices to generate for each input message.

To see all the parameters available, go to the official documentation.

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